Yu Ding

Postdoctoral Research Fellow ▪ Machine Learning for Healthcare ▪ Cancer Data Science

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Research Profile

Statistical machine learning researcher developing methods for heterogeneous biomedical data, with a focus on cancer evolution and intratumor heterogeneity. Research spans scalable subclonal reconstruction, multi-omics integration, structured statistical modeling, and privacy-preserving functional learning.

Academic and Research Appointments

The University of Texas MD Anderson Cancer Center — Houston, Texas
Postdoctoral Research Fellow — 2024 – Present
Department of Bioinformatics and Computational Biology; mentor: Wenyi Wang.

The University of Kansas School of Business — Lawrence, Kansas
Research Assistant in Operations Management — 2019 – 2020

Education

Binghamton University — Vestal, New York
Ph.D. in Industrial and Systems Engineering — 2020 – 2024
Advisor: Bing Si.
Dissertation: *Statistical Machine Learning and Data Fusion Methodologies: Applications in Healthcare*.

Wayne State University — Detroit, Michigan
M.S. in Industrial Engineering — 2017 – 2019

University of Science and Technology of China — Hefei, China
B.S. in Geophysics — 2013 – 2017

Peer-Reviewed Publications

†: Equal first author *: Corresponding author

[J.8] Mueller, S. D.*, Sutherland, M. A., **Ding, Y.**, Si, B., Hutchinson, M. K. (2026). **A cross-sectional analysis of comprehensive campus health services at MSI and non-MSI institutions in the U.S.** *Journal of Health Equity*, 3(1), 2679111.

[J.7] **Ding, Y.**, Costa, C., Si, B.* (2026). **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**. *IEEE Transactions on Automation Science and Engineering*, 23, 4704–4716. [Video]

[J.6] **Ding, Y.**, Somers, V. K., Si, B.* (2025). **Multi-modal mixed-type structural equation modeling with structured sparsity for subgroup discovery from heterogeneous health data**. *IISE Transactions*, 57(12), 1497–1511.

[J.5] Sutherland, M. A.*, Hutchinson, M. K., Si, B., **Ding, Y.**, Liebermann, E., Connolly, S. L., Saint-Eloi Cadely, H., Hayes, J. F., Mueller, S. D. (2025). **Health screenings in college health centers: Variations in practice**. *Journal of American College Health*, 73(8), 3031–3038.

[J.4] Mueller, S. D.*, Sutherland, M. A., Hutchinson, M. K., Si, B., **Ding, Y.**, Connolly, S. L. (2024). **Student Health Services at Historically Black Colleges and Universities and Predominantly Black Institutions in the United States**. *Health Equity*, 8(1), 226–234.

[J.3] Alramadeen, W., **Ding, Y.**, Costa, C., Si, B.* (2023). **A novel sparse linear mixed model for multi-source mixed-frequency data fusion in telemedicine**. *IISE Transactions on Healthcare Systems Engineering*, 13(3), 215–225.

[J.2] Jiang, L., **Ding, Y.**, Sutherland, M. A., Hutchinson, M. K., Zhang, C., Si, B.* (2022). **A novel sparse model-based algorithm to cluster categorical data for improved health screening and public health promotion**. *IISE Transactions on Healthcare Systems Engineering*, 12(2), 137–149.

[J.1] **Ding, Y.**, Yang, Q.*, King, C. B., Hong, Y. (2019). **A general accelerated destructive degradation testing model for reliability analysis**. *IEEE Transactions on Reliability*, 68(4), 1272–1282.

Preprints and Other Manuscripts

†: Equal first author *: Corresponding author

Preprints

[P.2] Jiang, Y.†, Montierth, M. D.†, **Ding, Y.**, Yu, K., Tran, Q., Wu, A., Li, R., Ji, S., Liu, X., Shin, S. J., Cao, S., Tang, Y., Lesluyes, T., Kimmel, M., Wang, J. R., Tarabichi, M., Zhu, H., Van Loo, P., Wang, W.* (2026). **Scalable subclonal reconstruction of cancer cells in DNA sequencing data using a penalized likelihood model**. *bioRxiv* preprint, version 2, May 7, 2026.

[P.1] Paul, A.†, Lal, J. C.†, Ji, S., Fong, C., Chen, K., **Ding, Y.**, Li, R., Dai, Y., Tran, Q., Montierth, M., Alberti, S., Kopetz, S., Wang, W.* (2026). **Tumor cell specific total mRNA expression informed neural networks predicts cancer progression**. *bioRxiv* preprint, version 1, May 6, 2026.

Manuscripts under review

[M.4] Shalapour, S.*, et al. (including **Ding, Y.**). **Host Nutritional State Reprograms Hepatic Anti-Tumor Immunity, Shaping Tumor Progression and Immunotherapy Outcome**. Manuscript under review.

[M.3] Shalapour, S.*, et al. (including **Ding, Y.**). **Macronutrient Composition and Alcohol Drives Distinct Molecular and Oncogenic Trajectories in the Liver**. Manuscript under review.

[M.2] **Ding, Y.**, Costa, C., Si, B.* **Single-Shot Differentially Private Vertical Federated Function-on-function Learning**. Manuscript under review. [Code]

Manuscript in preparation

[M.1] Dai, Y., Guo, S., Tran, Q., Zhang, Z., **Ding, Y.**, Hosseini, N., Basu, S., Cruz-Colli, C., Subudhi, S., Goswami, S., Bose, R., Soundararajan, R., Haffner, M., Van Loo, P., Aparicio, A. M.*, Wang, W.* **Multimodal dissection of metastatic tumors identifies *TP53* truncation as a determinant of chemotherapy resistance in Aggressive Variant Prostate Cancers**. In preparation.

Research Software

- **CliPP** — contributor. Penalized-likelihood reconstruction of tumor subclones from single-sample DNA sequencing data; preprocessing and postprocessing updates. [Repository] [Release]
- **CliPP1.5** — developer and maintainer. Single-sample mutation clustering with estimates of cancer cell fractions and mutation multiplicities; CPU and GPU support. [Repository]
- **CliPP2** — developer and maintainer. Joint mutation clustering across tumor regions, estimating regional cancer cell fractions and mutation multiplicities. [Repository]

Honors, Awards, and Fellowships

Warren Alpert Computational Biology and AI Network Fellow; *UCLA Computational Genomics Summer Institute* — Aug 2025

Best Team Award; *James P. Allison Institute and IDSO Hackathon* — Apr 2025

QCRE Best Student Paper Competition Finalist; *IISE Annual Conference & Expo* — May 2024

National Science Foundation (NSF) Travel Award; *The University of Arizona* — May 2024

Watson Professional Development Fund; *SUNY Binghamton* — Apr 2024

Summer Research Fellowship; *The University of Kansas* — Apr 2020

University Graduate Fellowship; *The University of Kansas* — Aug 2019

Graduate Fellowship; *Wayne State University* — Aug 2018

National Encouragement Scholarship; *University of Science and Technology of China* — Aug 2016

Talks and Posters

Talks and conference presentations

- **Scalable subclonal reconstruction of cancer cells in DNA sequencing data using a penalized likelihood model**. Poster, NCI Summer School on Algorithmic Cancer Biology (SSACB), National Institutes of Health, Bethesda, MD, July 6–10, 2026. [Poster]
- **Bidirectional Prognostic Associations of Subclonal Mutation Fraction in Patients Receiving Immunotherapy**. Oral presentation, UCLA Computational Genomics Summer Institute (CGSI), UCLA, Los Angeles, CA, 2026.
- **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**. Departmental talk, Department of Radiation Oncology, Mayo Clinic Arizona, Phoenix, AZ, July 1, 2024.
- **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**. Conference presentation, IISE Annual Conference & Expo, Montreal, Canada, May 18, 2024. [Video]

Coauthored posters

- Jahedi, A., Unal, M., **Ding, Y.**, Nurmammadova, A., Liu, Y., Wang, L., Nolan, G. P. **Boundary-Specific Remodeling Defines Divergent TLS Maturation in Colorectal Cancer**. James P. Allison Institute 3rd Annual Scientific Symposium, MD Anderson Cancer Center, Houston, TX, October 24, 2025. [Poster]
- Nurmammadova, A., **Ding, Y.**, Jahedi, A., Unal, M., Liu, Y., Wang, L., Nolan, G. P. **Deep Learning–Based Spatial Analysis of TLSs in CRC Reveals Stromal Barriers Predicting Impaired Cell–Cell Communication**. James P. Allison Institute 3rd Annual Scientific Symposium, MD Anderson Cancer Center, Houston, TX, October 24, 2025. [Poster]

Teaching and Mentoring

Guest Lecturer — Spring 2025
Rice University — Lecture recording

- STAT 623/423: Probability in Bioinformatics and Genetics (lectures, review sessions, exam questions, office hours)

Student Mentor
MD Anderson Cancer Center

- Mentoring Ph.D. rotation students and laboratory interns.

Professional Service and Memberships

- **Journal Reviewer:** *Nature Reviews Cancer*; *Nature Genetics*; *Bioinformatics*; *Journal of the American Statistical Association*; *IEEE Transactions on Medical Imaging*.
- **Memberships:** Institute of Industrial and Systems Engineers (IISE); Institute for Operations Research and the Management Sciences (INFORMS); Institute of Electrical and Electronics Engineers (IEEE).